

LAIZHEN LI

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EDUCATION

University of Chinese Academy of Sciences, China
Master of Engineering in Computer Technology
Advisor: Xitong Gao

Sep. 2024–Present (Expected Jul. 2027)
GPA: 3.71/4.0

Henan University of Engineering, China
Bachelor of Engineering in Software Engineering

Sep. 2019–Jul. 2023
Average: 81/100

RESEARCH INTERESTS

I study how to make large language models (LLMs) robust to adversarial misuse, how to secure LLM agents operating in untrusted environments, and how to improve the effectiveness and efficiency of agent systems.

RESEARCH EXPERIENCE

Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences

Jun. 2025–Present

Master's Student

Advisor: Xitong Gao

LLM Jailbreak Defense — StreamGuard

- Designed and implemented **StreamGuard**, a lightweight, plug-and-play decoding-time defense against jailbreak attacks that self-reviews partial generations and inserts safety prompts when unsafe content is detected, without fine-tuning.
- Evaluated StreamGuard on Llama-3.1-8B-Instruct and Qwen2.5-7B-Instruct against AutoDAN, GCG, and DeepInception; achieved a mean ASR of **1.7%**, compared with **20.3%** for the best-performing defense baseline and **60.3%** without defense, across six model–attack settings.

MCP Agent Security — A2M

- Designed and implemented **A2M (Attraction to Manipulation)**, a two-stage black-box attack that optimizes tool meta-data (tool names and descriptions) to increase selection likelihood, then refines return payloads using execution traces to manipulate downstream reasoning.
- Evaluated on LiveMCPBench across five frontier models; when optimized directly against GLM-4.6, A2M achieved a mean ASR of **74.4%** (64.3–92.9%) across three ASR-based objectives (information exfiltration, unauthorized environment modification, and reasoning derailment) and increased token consumption by **32.4×** on a fourth, token-wasting objective; the optimized attacks remained effective when transferred to four unseen models.

Efficient LLM Agents — Growing Harness

- Designed and implemented **Growing Harness**, a failure-guided framework that starts from a strategy-free scaffold and incrementally grows it into an executable agent harness through function-level execution traces and localized code edits, shifting recurring control logic from LLM inference to reusable program code.
- Evaluated on BrowseComp-Plus and WebArena-Verified across three backbone models; achieved a mean task success rate of **42.2%** across six benchmark–model settings, versus **23.6%** for the ReAct-style Tool-Calling baseline (**+18.6 pp**), while reducing LLM calls by **76.0–91.8%** and inference cost by **74.4–98.6%**.

PUBLICATIONS

[1] StreamGuard: A Streaming-based Defense Against Jailbreaking Attacks in Large Language Models
[Laizhen Li](#), Xitong Gao, Xuan Wang, Juanjuan Zhao, and Kejiang Ye.
ICONIP 2025. [Paper]

[2] JADE: Jailbreak-Aware Dynamic Defense Enhancer for LLMs via Reinforcement Learning
Peicheng Zhao, [Laizhen Li](#), Xitong Gao, Juanjuan Zhao, and Kejiang Ye.
IEEE BigData 2025. [Paper]

[3] A2M: Trace-Optimized Agent Hijacking in the MCP Ecosystem
[Laizhen Li](#), Xuan Wang, Peicheng Zhao, Juanjuan Zhao, Kejiang Ye, Cheng-zhong Xu, and Xitong Gao.
AACL-IJCNLP 2026, accepted.

MANUSCRIPTS UNDER REVIEW

[1] Grow the Harness, Not the Context: From Strategy-Free Scaffolds to Reusable Specialist Agents
[Laizhen Li](#), Jiarui Li, Juanjuan Zhao, Kejiang Ye, Ye Li, Cheng-zhong Xu, and Xitong Gao.
AAAI 2027. Under review.

[2] Co-Evolving Poison and Defense Harnesses: Adversarial Arms Races for Indirect Prompt Injection in Tool-Integrated Agents
Peicheng Zhao, [Laizhen Li](#), Juanjuan Zhao, Kejiang Ye, and Xitong Gao.
AAAI 2027. Under review.

SKILLS

Programming & Systems: Python, Bash, Linux, Git, Docker, CUDA.
Machine Learning: PyTorch, SFT, PPO, LoRA, Hugging Face Transformers, Weights & Biases.
LLM & Agent Systems: vLLM, LangChain, MCP.